

Statement of Purpose

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When I first used an old computer during my childhood, it was slow and struggled to run even simple programs. Instead of feeling frustrated, I became curious about why it was slow and how computers worked inside. That curiosity has since grown into a passion for building efficient computing hardware, from understanding microarchitectural trade-offs to creating optimized SoC and FPGA-based systems.

I have always been strongly interested in Mathematics, Computer Science, and Technology, and my curiosity to learn independently has played a major role in shaping who I am today. After completing my school education, I entered the **Faculty of Engineering at the University of Peradeniya**, one of Sri Lanka's most reputed universities, by achieving an **island rank of 149** out of 19,508 candidates in the Physical Science stream at the G.C.E. Advanced Level Examination, one of the country's most competitive exams. I completed my **B.Sc. Engineering (Hons.) in Computer Engineering** with a **perfect GPA of 4.00/4.00**, first in a cohort of 486 students. My academic journey has been a continuous exploration of how systems work from the inside out and how small improvements can make a big difference in performance.

During my second year, I was introduced to **Computer Architecture**, which quickly became one of the most inspiring subjects in my studies. It gave me a structured understanding of how processors, caches, and memory hierarchies interact to make a system efficient. I was fascinated by how small architectural design choices could dramatically affect system performance and power behaviour. When it came time to select electives, I chose Advanced Computer Architecture and Advanced Embedded Systems, as well as Image Processing, Neural Networks, and Artificial Intelligence out of curiosity. These courses provided the foundation to understand intelligent systems and modern computing, and helped me connect hardware design concepts with the growing field of machine intelligence.

My first exposure to processor design came through building an [8-bit single-cycle processor](#) in Verilog HDL, featuring instruction and data caches, a memory unit, and a MIPS-inspired ISA supporting arithmetic, logic, and control operations. I developed a comprehensive testbench for functional verification and waveform analysis using GTKWave, which strengthened my understanding of datapath and control logic design. I later worked on a **JPEG encoding optimization project** for an Advanced Embedded Systems coursework, where we developed a **pipelined multiprocessor system-on-chip (MPSoC)** on an Altera Cyclone IV FPGA to accelerate image compression. After analyzing performance bottlenecks, particularly in the Discrete Cosine Transform (DCT) stage, I implemented **custom hardware instructions** to perform the DCT directly in hardware. These optimizations significantly improved throughput, and our work was **published at [IEEE ICAC 2024](#)**. This project was a key step in developing my ability to identify performance issues and resolve them through architectural and hardware-level improvements.

Building on that, my **final-year project (SNAP-V)** became the most defining part of my undergraduate journey. Together with my team, I designed and developed a **dual-core RISC-V System-on-Chip integrating a configurable neuromorphic accelerator with over 1,024 LIF neurons** organized into parallel clusters interconnected through a hierarchical Network-on-Chip (H-NoC). This architecture enabled efficient spike-based computation for small-scale spiking neural networks targeting low-power edge applications such as robotics. We **validated the SoC on the MNIST dataset using Synopsys VCS/PrimePower and Xilinx Vivado**, achieving **96.69% accuracy (within 2.62% of baseline) and a state-of-the-art energy efficiency of 1.39 pJ per synaptic operation**. The project gave me end-to-end experience in RTL development, ASIC design methodologies, verification, synthesis, and detailed power/performance analysis. Through this, I gained a clearer understanding of how hardware designs transition from behavioral RTL to real silicon implementations. It also deepened my understanding of processor–accelerator integration and workload characterization in FPGA-based systems. Currently, under the **PeraMorphIQ Neuromorphic Research Group**, I continue extending this work while mentoring junior teams on on-chip learning integration, power optimization, and verification workflows for next-generation accelerators.

To further strengthen my microarchitectural foundation, I implemented a [5-stage pipelined RISC-V RV32IM processor with in-order hazard handling](#). Through this project, I developed a concrete understanding of how pipeline stages interact, how timing delays propagate across stages, and where bottlenecks typically emerge in a real processor pipeline. During RTL-level power and static timing analysis using Synopsys tools, I identified that the ALU contributed disproportionately to total power consumption compared to the register file. After implementing targeted optimizations, we successfully reduced ALU power, improving the processor's overall efficiency. The final design achieved **0.197 mW at 142 MHz**, and I automated the entire analysis workflow through **GitHub Actions CI/CD**. The design was prototyped on an **Xilinx Virtex-7 FPGA** for real-time validation. I am now extending this work with a spiking neural network–based branch prediction mechanism, exploring biologically inspired approaches to improve prediction accuracy and energy efficiency. This project bridges traditional pipelined CPU design with adaptive, event-driven computation, reflecting my interest in efficient and intelligent hardware systems.

Alongside projects, I gained industry experience during my internship at [WSO2](#), where I developed Ballerina integrations, including the OpenAI Finetunes Connector and ISO20022-to-SwiftMT conversion modules. This experience deepened my understanding of large-scale system design and software–hardware interoperability. Currently, as a [Temporary Instructor](#) in the Department of Computer Engineering, I conduct Computer Architecture and Embedded Systems laboratory sessions, mentor project teams, and assist with evaluations, an experience that has strengthened my ability to communicate and collaborate in research-driven environments.

In the future, I hope to contribute to designing hardware systems where performance, efficiency, and intelligence converge—building adaptive architectures that power the next generation of computing across robotics, edge, and embedded platforms. For me, engineering is about continuous innovation: understanding systems deeply and improving them to shape a smarter, more efficient tomorrow.